



AI Playbook and Toolkit for Public Health Departments

Introduction to AI for Public Health

INT 100

Artificial intelligence is a broad family of computational methods that can recognize patterns, generate content, support prediction, retrieve information, or assist with multi-step workflows. In public health, AI is not a single product or a single capability. It may appear inside analytics platforms, dashboards, document tools, translation systems, vendor products, chatbots, surveillance workflows, or administrative systems. This module gives you the vocabulary and judgment needed to participate in AI discussions before a tool is selected or a pilot begins. The goal is not to turn every learner into a data scientist. The goal is to help you recognize different kinds of AI, ask the right implementation questions, and understand why governance, privacy, equity, security, and human review are necessary. You should use this module before completing the readiness assessment, reviewing vendors, prioritizing use cases, or participating in governance discussions. A shared foundation helps prevent confusion between predictive models, generative tools, retrieval systems, deep research, and agentic workflows.

Level	Foundational Module
Primary track	Foundational Modules
Appears in tracks	shared-foundational

Learning Objectives

- Define artificial intelligence in practical public health terms.
- Distinguish predictive AI, generative AI, retrieval-augmented generation, deep research, and agentic AI.
- Explain why AI outputs require human review, governance, privacy protections, and equity review.
- Identify appropriate and inappropriate uses of AI in public health workflows.
- Describe the first questions to ask before an AI tool is used, procured, piloted, or deployed.

Module Overview

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Definitions

- Artificial intelligence (AI): A broad set of computational methods that can recognize patterns, generate content, support prediction, or assist action toward defined goals.
- Predictive AI: AI that uses data patterns to estimate a future event, risk, classification, or outcome.
- Generative AI: AI that creates new text, images, code, summaries, translations, or other content from prompts and source material.
- Retrieval-augmented generation (RAG): A method that connects a generative AI system to approved source materials so responses can be grounded in specific documents or knowledge bases.
- Deep research: An AI-assisted research workflow that plans searches, reviews sources, synthesizes findings, and produces a structured report for human validation.
- Agentic AI: AI that can coordinate multiple steps or actions within defined boundaries, often across tools or systems, while still requiring human oversight.
- Human-in-the-loop: A workflow design in which a qualified person reviews, approves, corrects, or rejects AI outputs before consequential action occurs.

Jurisdiction and Agency Policy Note

AI use must be interpreted within your own jurisdiction's legal, policy, privacy, procurement, public records, labor, civil rights, and data governance requirements. A use that is appropriate in one health department may require additional review or may be prohibited in another because of data sensitivity, statutory authority, vendor terms, local policy, or community context.

If your department already has data governance, technology governance, privacy review, cybersecurity review, records management, procurement, accessibility, equity review, or quality improvement processes, AI governance should leverage those structures whenever possible. The playbook and tools are meant to fill gaps, not replace effective processes that already exist.

Public health agencies make decisions that affect communities, services, communications, surveillance, emergency response, and public trust. AI can support that work, but it can also introduce errors, bias, privacy exposure, automation over-trust, or confusion about accountability if it is adopted without a shared understanding.

AI systems are often described with broad marketing language that hides important differences between tools. A chatbot that drafts plain-language material, a predictive model that estimates outbreak risk, a retrieval system that searches internal policy, and an agentic workflow that creates tasks across systems all require different safeguards.

Understanding these differences helps you participate in responsible use case selection. It also helps you recognize when a proposed AI use belongs in learning and exploration, when it needs formal governance review, and when it should not proceed.

A preparedness team wants to use AI to help summarize after-action reports and identify recurring improvement themes. A generative AI tool might draft summaries from approved documents, while a retrieval system could point reviewers back to the relevant source passages. The team would still need a human reviewer to verify accuracy, remove sensitive information, identify local context, and decide which findings should become improvement actions.

The same workflow could become riskier if staff entered confidential incident details into an unapproved public tool or accepted the summary without checking the source documents. This example shows why the first AI question is not simply whether the tool can produce a useful draft. The first question is whether the workflow, data, review process, and accountability structure are appropriate.

AI should be understood as a set of capabilities rather than a single technology. Predictive AI estimates patterns or outcomes, generative AI creates content, retrieval systems connect responses to source documents, deep research supports structured evidence gathering, and agentic AI coordinates steps in a workflow. Each capability can be useful, but each creates different risks and review needs.

The public health workflow matters as much as the model. A low-risk drafting task using public information is different from a workflow that influences inspections, benefits, outbreak response, case prioritization, or public alerts. You should always identify the workflow step, the user, the people affected, the data involved, and the decision that remains human.

Human review is not a vague concept. A responsible workflow names who reviews the AI output, what evidence they check, what criteria they apply, how corrections are documented, and when concerns are escalated. Review should be built into the workflow before the tool is used.

Deep research tools deserve special attention because they can look like expert evidence synthesis. They may be useful for finding, organizing, and summarizing sources, but the learner or reviewer must still check source quality, missing perspectives, publication dates, jurisdictional relevance, and whether the conclusions are supported.

AI systems can generate inaccurate, incomplete, biased, outdated, or misleading outputs. Generative tools can hallucinate. Predictive tools can perform poorly for underrepresented groups. Agentic systems can take inappropriate actions if boundaries, permissions, or escalation paths are unclear.

Public health data can be sensitive even when it is not formally protected health information. Records may include small populations, geographic identifiers, outbreak details, program participation, immigration-related concerns, tribal data, environmental exposure information, or other context that requires careful handling.

Strong guardrails include approved-use policies, data tiering, human review, source verification, equity review, access controls, documentation, incident reporting, and monitoring. These safeguards should be proportional to risk, but they should not be skipped because a tool appears easy to use.

When you encounter a proposed AI use, start by describing the public health task in plain language. Identify the problem, the current workflow, the proposed AI-supported step, the users, the data, the affected populations, and the decision points that must remain under human control.

Next, decide what type of AI is being proposed. Predictive, generative, retrieval, deep research, and agentic uses should not be reviewed as if they were the same. The review pathway should match the risk, data sensitivity, workflow impact, and degree of automation.

Finally, connect the proposed use to the playbook. Early ideas may belong in vision, readiness, governance, stakeholder engagement, or use case intake before any pilot or procurement begins. The tools are available to fill gaps where your department does not already have a comparable process.

What AI tools or AI-enabled features are already being used informally in your work environment?

Which public health workflows in your area involve drafting, summarizing, searching, triage, prediction, routing, or monitoring tasks that could attract AI proposals?

What information should never be entered into an unapproved AI tool in your role?

Who would need to review an AI output before it influenced communication, services, enforcement, surveillance, or leadership decisions?

What existing governance, privacy, security, procurement, or equity review process could be used before creating a new AI-specific process?

Choose one public health workflow where AI might be proposed. Write a short description of the current workflow, the AI-supported step, the type of AI that might be involved, the data that would be used, the staff who would use the output, and the people or communities who could be affected.

Then identify at least three safeguards that would be needed before the AI-supported workflow could move forward. Your safeguards should address human review, data protection, equity, accuracy, documentation, governance review, or monitoring.

Upload or paste a one- to two-page AI workflow concept note.

The concept note should identify the proposed AI type, workflow step, data involved, users, affected populations, human review point, and key risks.

It should also list the governance, privacy, equity, security, procurement, or leadership questions that need to be answered before the idea moves forward.

Use AI-assisted feedback to help you identify strengths, missing elements, unclear assumptions, and areas that may need revision.

Expected Assignment

- A short description of a public health workflow where AI might be used.
- A list of the data, users, affected populations, and human decision points in that workflow.

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- A preliminary note describing why the use case may need governance, privacy, equity, security, procurement, or leadership review.
 - A reflection on where AI should not be used or where human judgment must remain primary.

Knowledge Check

- 1. What is the most important first step after completing the Introduction to AI for Public Health module?
- 2. Which practice best supports responsible AI use in public health?
- 3. Which statement best reflects the risk or guardrail for this module: AI systems can generate inaccurate, incomplete, biased, outdated, or misleading outputs. Generative tools can hallucinate. Predictive tools can perform poorly for underrepresented groups. Agentic systems can take inappropriate actions if boundaries, permissions, or escalation paths are unclear.
- Public health data can be sensitive even when it is not formally protected health information. Records may include small populations, geographic identifiers, outbreak details, program participation, immigration-related concerns, tribal data, environmental exposure information, or other context that requires careful handling.
- Strong guardrails include approved-use policies, data tiering, human review, source verification, equity review, access controls, documentation, incident reporting, and monitoring. These safeguards should be proportional to risk, but they should not be skipped because a tool appears easy to use.
- 4. When should a staff member escalate an AI-related concern?
- 5. What counts as stronger completion evidence for a training module?

References and Resources

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